UNBLIND

# Build plan for Danny

Pre-dosing module, master agent and mission control (platform lead)

**Your mission tonight:** Make the trial start right, and make the whole platform run as one product.

**Why this is yours:** You are the strongest on agent development and front and back end. Pre-dosing is the most product-shaped module (a protocol editor, an applicant portal, a screening funnel), and the master agent plus mission control is where the three modules become one product. Callum and Izzy own the modules that need the most statistics and domain depth.

| At a glance | |
|---|---|
| Deadline | Working prototype on main by 00:00 tonight. Write-up, pitch and rehearsal tomorrow. |
| Your teammates | Callum and Izzy build the other two modules in parallel. You only meet through the shared contract. |
| How to read this | Sections 1 to 5 are the same in all three plans: the whole project. Your personal sections follow, then the shared timeline, demo script and glossary. |
| Golden rules | Contract frozen at 20:00. Own your folder. Every agent works with no API key. Every decision is logged. Every trial-changing action goes to a human. Synthetic data is labelled. |

# 1. The project in one page

**UNBLIND is an agentic platform that runs a psychedelic clinical trial with a team of AI agents, from protocol design to the FDA report, so the trial measures and controls the bias that sinks these drugs today.**

### The problem in one breath

To prove a drug works, you compare a drug group with a placebo group and nobody knows who got what. Psychedelics break this: patients know they tripped, and so do the clinicians scoring them. Every result becomes drug effect plus expectation, stuck together. The FDA's final psychedelic guidance (July 2026) names this functional unblinding as the central threat to validity, and it was central to the 2024 rejection of the MDMA application.

### Why it stays unsolved

Nobody owns blinding as a job. Most trials never measure it, the ones that do measure it differently, and problems are only discovered at the end, after the money is spent. Trials are also slow to fill, and screening volunteers is manual.

### What we are building

A master agent coordinates three modules, one per phase of a trial. Each module is a small team of agents that does real work every simulated day and brings decisions to a named human.

| Phase | What the agents do | Owner |
|---|---|---|
| Pre-dosing | Co-design the protocol (risk, fixes, drafted blinding section, re-check on every edit), write ethics-ready recruitment ads, pre-screen applicants automatically, build the standard measurement plan and schedule | Danny |
| Dosing (acute) | Run post-session check-ins (guess, confidence, expectancy), structure monitor debriefs, monitor blinding per site and per rater through a sealed vault, detect drift, propose actions, chase missing data, scan new literature weekly | Callum |
| Post-dosing | Standard follow-up collection, a researcher question portal with a bias gate, a harmonised data export, and the FDA blinding report with an expectancy-adjusted effect | Izzy |
| Master agent | Runs the clock, routes messages between agents, keeps the trial memory, queues human decisions, shows mission control | Danny |

### What the judges will see

- A judge pastes or edits a draft protocol. The pre-dosing agents score its blinding risk, fix it, and screen 260 synthetic applicants down to a clinician-approved shortlist.
- Press play. Simulated weeks pass in about two minutes: sessions, check-ins, debriefs and a live blinding chart per site.
- Halfway through, raters at one site start guessing right far too often. The dosing agents spot it, explain it and propose moving those raters to remote assessment. The judge clicks approve.
- The judge can play a patient and answer a check-in.
- At close-out the post-dosing agents produce a standard FDA blinding report: every measurement, alert and human decision, plus the reported and expectancy-adjusted effect.

Everything simulated is SYNTHETIC and labelled as such, which the hackathon rules allow. No real patient data, ever.

# 2. The bottlenecks, quantified and sense-checked

The brief says every claim must name a bottleneck, quantify it, and show exactly how we move it. These are the numbers we can stand behind tonight. Anything not listed here needs a source before it goes in the pitch.

| Bottleneck | Number (with source) | How UNBLIND moves it | Metric we report |
|---|---|---|---|
| Blinding is rarely measured | Only 29.5% (33 of 112) randomised psychedelic trials assessed blinding (systematic review, JAMA Psychiatry 2026) | Every participant and rater gets the same scheduled guess check-in by default | Share of scheduled blinding check-ins completed |
| When measured, blinding fails | Psilocybin, LSD and ayahuasca studies frequently reported failure above 90%; inert-placebo MDMA trials above 85% (same review) | Risk predicted before the trial; drift caught per site and per rater within days | Days from drift onset to alert |
| Regulators now demand it | FDA final guidance, July 2026: blinding questionnaires, expectancy measures before randomisation and at end of treatment, central blinded raters, alternative controls | Protocol checked against each item; the evidence collected into one report | Guidance items satisfied, before and after |
| Sessions are labour-heavy | Two monitors per session and a physician within 15 minutes (FDA guidance) | Debriefs spoken or typed, structured automatically into adverse events and unblinding cues | Debriefs structured automatically |
| Effects are small | COMP005: -3.6 MADRS vs placebo (95% CI -5.7 to -1.5). COMP006: -3.8 vs 1 mg psilocybin (95% CI -5.8 to -1.8) (Compass press releases) | Close-out report shows reported and expectancy-adjusted effect side by side | Reported vs adjusted effect, with interval |
| Blinding data is not comparable | Of 18 real trials our research agent examined, only 4 report clean per-arm guess data; timing ranged from dosing day to day 365 (our candidate gold set, unverified) | One standard instrument set, UNBLIND-BLIND-1, at fixed timepoints, exported in one schema | Trials exporting in the standard schema |
| Recruitment and screening are slow and manual | NEEDS A SOURCE. Do not quote a number until Callum or Izzy finds one | Agent screens every applicant in seconds; clinicians review only a shortlist and the flagged cases | Share screened automatically; clinician reviews per enrolled participant; staff hours saved (state the per-applicant minutes as an assumption) |

## Sense checks: what a scientist judge will push on

- **Bigger trials do not reduce bias.** More patients shrink random error, not the systematic error from knowing your group. A 1,000-person trial with broken blinding is just precisely wrong. Pitch recruitment as faster and fairer, not as less biased. Also, phase 3 psychedelic trials are already large (COMP006 dosed 581), so say 'most academic phase 2 trials are small'.
- **Pre-screening cannot remove humans entirely.** A clinician legally confirms final eligibility, especially for psychosis history, heart conditions and medication. The agent automates the funnel so humans review only a shortlist. That is the honest efficiency claim.
- **Prior psychedelic use: balance it, do not exclude it.** Experienced users recognise the drug faster, but excluding them shrinks the pool. Stratified randomisation spreads them evenly across arms.

- **Targeted ads using health or drug-use interests are risky** under UK GDPR (special category data) and are restricted by ad platforms. We park targeting and build compliant, ethics-ready ads instead.
- **Researcher questions can themselves break blinding.** Asking about 'mystical experiences' before the primary result is locked tells patients what they got. Every question passes a bias gate and is released only after the primary lock.
- **Reverse causation.** Patients who improve may guess 'drug' because they improved, so correcting for guesses can remove real effect. The adjusted effect is a conservative bound, not the truth. Our synthetic test shows the size of this failure (about 0.15 g over-correction).
- **Facts to say correctly:** the 3.8 is COMP006 against 1 mg psilocybin, not placebo. 'Above 90% failure' applies to many psilocybin, LSD and ayahuasca studies, not to all trials. Compass has not been 'cleared': rolling review is a submission route, not a verdict.
- **Prior art:** Szigeti and colleagues published an 'equal unblinding' meta-analysis in 2026. Our novelty is the agentic, living, design-time platform, not the idea that blinding matters.

# 3. How the platform works

## The shared foundations (the only things modules share)

| Piece | What it is | Why it matters |
|---|---|---|
| TrialRecord | One JSON document per trial holding the protocol, applicants, schedule, check-ins, metrics, alerts, approvals and events. Each module writes only its own sections. | Lets three people build in parallel without touching each other's code |
| MessageBus | Agents post messages (for example 'rater_drift'). The master agent routes them to whoever needs to know. | Agents cooperate without importing each other |
| ApprovalQueue | Any action that changes the trial becomes a request. A named human approves or rejects. Everything is logged. | The brief's 'human signs off key decisions', made concrete |
| Vault | The only holder of who got the drug. Agents receive group-level statistics only, with groups under 5 suppressed. Unsealing needs a named person and is logged. | Security: the AI cannot leak what it never sees |
| World | A SYNTHETIC simulator of applicants, patients, sites and raters, with planted events (rater drift at Site B from day 28, missing data at Site C, exclusions hidden in free text). | Lets us demo months in minutes, and evaluate agents against a known truth |

## The master agent

- Advances the simulated clock one day at a time and sets the phase: design, dosing, follow-up, close-out.
- Delivers messages and decides who hears about what. For example, rater drift goes to pre-dosing (draft a protocol amendment) and to post-dosing (record it in the FDA report). With a Gemini key it reads the message and chooses recipients; without one it uses a routing table. Either way the choice is logged.
- Isolates failures: if one module crashes, the others keep running and the error appears in the timeline.

- Writes a weekly summary and powers mission control: phase bar, timeline, approval inbox, agent message map, KPIs.

### The simulated timeline

| Day | Phase | What happens | Module |
|---|---|---|---|
| Setup | Design and recruitment | Protocol parsed and reviewed, fixes proposed, ads written, 260 applicants screened, clinician approves shortlist, stratified randomisation in the vault, measurement plan and schedule built | Pre-dosing |
| 1 to 20 | Dosing | Staggered dosing sessions; monitor debrief the same day; participant check-in the next day | Dosing |
| Each participant's weeks 1, 3, 6 | Dosing | Rater visit: MADRS score plus rater guess; participant check-in | Dosing |
| Daily | Dosing | Blinding metrics per site and rater, drift detection, missing-data chasing | Dosing |
| Weekly | All | Literature scan with human approval; master weekly summary | Dosing and master |
| After last week-6 visit | Primary lock | Dosing tells the master the primary endpoint is locked | Dosing to master |
| Weeks 8 and 12 | Follow-up | Standard follow-up collection; researcher questions released through the bias gate | Post-dosing |
| Day 90 | Close-out | FDA blinding report and harmonised export, signed off by a human | Post-dosing |

### What already exists and gets reused

- **The UNBLIND research pipeline** (already in the repo): eight agents that read any trial and extract its facts with verified quotes, a deterministic quote auditor, the meta-regression, the design advisor, a Streamlit app, an evaluation suite and 18 candidate real trials. In the platform it becomes the weekly literature agent (dosing) and the risk engine (pre-dosing and post-dosing).
- **A starter scaffold** drafted by Claude in its own workspace: core.py (record, bus, approvals, vault), world.py (simulator), master/orchestrator.py and MODULE_CONTRACT.md. It is not in the repo yet. Danny decides at kick-off whether to pull it in; Claude can sync it on request.

## 4. How this scores against the judging criteria

| Criterion | What a top score needs | Where UNBLIND delivers it | Owner |
|---|---|---|---|
| Uniqueness | Beyond a literature chatbot or stock model | Agents that live inside a trial for months; a vault the AI cannot read; a bias gate on researcher questions | All |
| Real-world need | One specific bottleneck, named user, plausible next step | Unmeasured, failing blinding. User: the clinical development lead running a psychedelic trial. Next step: pilot the standard kit with one contract research organisation | Izzy (narrative) |
| Science, data and evaluation | Tested against a baseline, data referenced and licence-checked, uncertainty stated | Drift detection vs end-of-trial-only checking; screening vs synthetic ground truth; adjusted vs naive effect on planted-truth simulations; sources and licences listed | Callum and Izzy |

| Criterion | What a top score needs | Where UNBLIND delivers it | Owner |
|---|---|---|---|
| Agent design and live demo | Real multi-step agent; demo handles a judge's new input | Judge edits a protocol, approves actions, plays a patient, submits a researcher question | Danny (shell), everyone (moments) |
| Security and responsible design | Named risks with mitigations; human signs off key decisions | Vault, approval queue, quote audit, bias gate, consent checks, GDPR notes, synthetic labelling | All |

# 5. Who does what

| Person | Owns | Mission |
|---|---|---|
| Danny | Pre-dosing module, master agent and mission control (platform lead) | Make the trial start right, and make the whole platform run as one product. |
| Callum | Dosing (acute) module: the agents that live inside the trial | Catch blinding failure while the trial can still fix it. |
| Izzy | Post-dosing module, the FDA blinding report and the science of the evaluation | Turn everything the trial collected into evidence a regulator and a scientist can trust, and make that evidence comparable across trials. |

**Danny, the next sections are yours.** Everything below is written for your module. Your teammates have the same shared sections and their own versions of these.

# 6. Your module: pre-dosing

## The bottlenecks you attack

- **Designs are locked before anyone predicts their blinding risk.** Sponsors pick comparator, rater setup and sample by convention, then learn at FDA review whether it was good enough. The FDA July 2026 guidance lists what it expects (blinding questionnaire, expectancy measures, central blinded raters, alternative controls). Your co-designer checks every item and drafts what is missing.
- **Screening is manual and slow.** Every applicant is checked by staff, often by phone. Your screening agent processes all applicants in seconds and sends clinicians only the shortlist and the flagged cases. Quantify it as: share screened automatically, clinician reviews per enrolled participant, staff hours saved (state minutes per manual screen as an assumption until Callum sources a figure).
- **Sense check to say out loud:** automation speeds recruitment, but bigger trials do not cure unblinding. And the final eligibility call stays with a clinician.

## Your agents

| Agent | What it decides (the agentic part) | Output | Human gate | No-key fallback |
|---|---|---|---|---|
| Protocol parser | Reads free-text protocols of any shape and decides drug, comparator type, rater model, prior-use policy, endpoint, sites, sample size, eligibility. Keeps the source sentence for each. | rec['protocol'] | None (read-only) | Regex and keyword rules |

| Agent | What it decides (the agentic part) | Output | Human gate | No-key fallback |
|---|---|---|---|---|
| Protocol co-designer | Predicts unblinding (u) and expected bias using the existing UNBLIND model; ranks fixes; runs the FDA checklist with evidence quotes; drafts the 'Blinding and expectancy' section. Re-checks automatically on every new protocol version. Reacts to 'rater_drift' messages by drafting an amendment. | rec['design_review'], rec['protocol_versions'] | Every fix and amendment is an approval request | unblind.model functions, rule checklist |
| Recruitment agent | Writes 3 ethics-ready ad variants (no promises, states placebo chance and time commitment, no sensational language), scores reading ease, picks consent-based channels. Parks behavioural targeting and says why (UK GDPR special category data). | rec['recruitment']['ads'] | Ads need ethics approval (state it) | Templates |
| Pre-screening agent | Applies the criteria table below; reads free text for disclosed conditions and never auto-includes them; assigns eligible, excluded or needs clinician, with reasons and quotes; checks fairness by age band and country. | rec['applicants'][i]['screening'], funnel counts | ONE approval: clinician confirms the shortlist and reviews flagged cases. Nothing is enrolled before that. | Rules plus regex on free text |
| Enrolment and randomisation | Enrols up to the target, sets strata (site x naive or experienced), calls vault.randomise, staggers dosing days 1 to 20. | rec['enrolled'], message 'enrolment_complete' | Follows the clinician approval | Deterministic |
| Measurement planner | Builds the standard UNBLIND-BLIND-1 plan and every schedule entry, with owners. | rec['measurement_plan'], rec['schedule'] | None | Deterministic |

### Pre-screening criteria (make these a visible, editable table)

| Criterion | Rule | Why |
|---|---|---|
| Age | 18 to 65 | Typical adult trial range |
| UK mobile | +44 7 number | Check-ins run through the UK chatbot |
| Distance | Near one of the sites | 8 in-person visits |
| Diagnosis | MDD or TRD; for TRD at least 2 failed antidepressants | Matches the protocol population |
| Psychosis, personal bipolar | Exclude | Safety; standard psychedelic exclusion |
| Uncontrolled cardiovascular condition | Exclude | Blood pressure rises during sessions |
| Pregnancy or breastfeeding | Exclude | Safety |
| Lithium | Exclude | Interaction risk |
| SSRI or SNRI | Needs clinician (washout decision) | Clinical judgement |
| Psychedelic use in last 12 months | Exclude | Recent exposure confounds |
| Earlier psychedelic use | Allowed, recorded for stratification | Balance, do not exclude |
| Free-text disclosure (bipolar, psychosis, cardiac, pregnancy, suicidal thoughts) | Needs clinician, with the quoted sentence | The agent must never auto-include a disclosed risk |
| Consent to chatbot check-ins | Required | The standard measurement depends on it |

## 7. The master agent and mission control

- Clock and phases: setup, then step one day, one week, or run to close-out. Auto-approve toggle for fast-forwarding, clearly labelled 'demo reviewer (auto)'.
- Message routing: routing table first, Gemini routing for unknown message kinds, every routing decision logged.
- Failure isolation: wrap every module call; show errors in the timeline instead of crashing.
- Mission control page: synthetic banner, phase bar, KPI strip across modules, timeline filtered by module and level, approval inbox (reviewer name required), agent message map (graphviz), links to each module page.
- Applicant portal: a form where a judge can apply as a volunteer and watch the screening agent decide live. This is your judge-input moment.

# 8. Your files and interfaces

| You own | You read | You send | You receive |
|---|---|---|---|
| unblind/platform/predose/*, master/*, core.py, world.py, app/platform_app.py, data/protocols/*, tests/platform/test_predose.py and test_master.py | Nothing from other modules; only the record sections they own, for display | protocol_amended, enrolment_complete | rater_drift, participant_unblinding (draft amendment), literature_update (re-check) |

Core requests you may get tonight: Izzy may ask for a Vault.effect_by_guess() aggregate for the close-out report. It is small: group outcomes by participant guess and arm, return means and counts with cells under 5 suppressed.

# 9. Your task list

| # | Time | Task | Priority |
|---|---|---|---|
| 1 | 19:45 | Set up branches; decide on and pull in the starter scaffold; freeze the contract | Must |
| 2 | 20:00 | Example protocols: a weak psilocybin TRD draft and a better MDMA PTSD draft (fictional, labelled) | Must |
| 3 | 20:10 | Protocol parser and co-designer (risk, fixes, checklist, drafted section, versioning, re-check) | Must |
| 4 | 20:50 | Pre-screening agent, clinician approval, enrolment, stratified randomisation | Must |
| 5 | 21:15 | Measurement planner and schedule (exact contract shapes; Callum and Izzy depend on it) | Must |
| 6 | 21:30 | Checkpoint: run Master end to end with whatever modules exist | Must |
| 7 | 21:45 | Mission control page and approval inbox | Must |
| 8 | 22:20 | Pre-dosing page: protocol editor, risk card, fixes with approve buttons, checklist, funnel, flagged applicants, fairness | Must |
| 9 | 22:45 | Applicant portal (judge applies live) | Should |
| 10 | 22:55 | Recruitment ads with reading ease | Should |
| 11 | 23:00 | Integration run x2, fix, freeze, tag | Must |
| 12 | later | Gemini routing and parsing; agent message map | Could |

# 10. Tests, evaluation and judge questions

- **Tests:** both example protocols parse; the checklist finds the planted gaps; free-text disclosures go to needs clinician; nothing enrols before approval; strata are balanced (vault counts only); a garbage protocol does not crash; Master runs setup to close-out with modules missing.
- **Evaluation numbers to report:** share screened automatically; free-text exclusions caught against the synthetic ground truth (evaluation only, never used by the agent); checklist items before and after fixes; predicted unblinding before and after approved fixes.
- **'Can an AI decide who joins a trial?'** No. It sorts and explains; a clinician confirms. Show the approval.

- **'Is targeting people by drug interest legal?'** We parked it for that reason; consent-based channels only.
- **'What if the model's risk number is wrong?'** It comes from a calibration that is synthetic tonight and labelled so; the checklist and fixes stand on the FDA guidance alone.

## 11. Tomorrow

- Lead the pitch deck and the 4-minute script.
- Lead three demo rehearsals, including one where someone throws an unexpected input.
- Tidy the README and repo; make sure the judge path works offline.

# 12. How we work in parallel tonight

- **Contract freeze at 20:00.** After that, nobody changes the shared shapes in MODULE_CONTRACT.md without telling the other two in the group chat.
- **Own your folder.** Danny: unblind/platform/predose/, master/, core.py, world.py and app/platform_app.py. Callum: unblind/platform/dosing/. Izzy: unblind/platform/postdose/ and eval/platform/. Nobody edits another person's folder.
- **Git.** One branch each (danny/predose, callum/dosing, izzy/postdose). Pull main at every checkpoint. Danny merges.
- **Every module works with no Gemini key.** Deterministic fallback first, Gemini second. The demo must survive bad Wi-Fi.
- **Every agent explains itself.** rec.event(agent, text, level, module) for each decision. Decisions are what make it look and behave like an agent.
- **Human gate.** Anything that changes the trial goes through the ApprovalQueue.
- **Vault only.** Never compute anything from allocation outside the vault.
- **Label synthetic data** on every page and in every chart.
- **Copy:** UK English, plain language, no em dashes.

## Tonight's timeline

| Time | Everyone | Checkpoint |
|---|---|---|
| 19:45 to 20:00 | Read this plan. Agree the contract. Danny sets up branches and the scaffold. | Contract frozen |
| 20:00 to 21:30 | Build block 1: agents with deterministic fallbacks, plus offline tests | Each module runs inside the master with the others present or missing |
| 21:30 to 21:45 | Checkpoint call: run Master end to end on main | Known bugs listed |
| 21:45 to 23:00 | Build block 2: module page, KPIs, Gemini enhancements, your demo moment | Pages render in mission control |
| 23:00 to 23:30 | Integration: Danny runs the full demo path twice; others fix their bugs | Clean run from setup to close-out |
| 23:30 to 00:00 | Freeze. One rehearsal of the demo path. Push and tag v1. | Working prototype on main |

## Tomorrow

| Task | Lead | Support |
|---|---|---|
| One-page write-up (problem, who it helps, data, evaluation, limits and safeguards) | Izzy | Callum (evaluation numbers), Danny (architecture) |
| Pitch deck and 4-minute script | Danny | Izzy (science slides), Callum (evaluation slide) |
| Judge Q&A sheet (sense checks above, with answers) | Callum | Izzy |
| Demo rehearsal x3, including a judge's unexpected input | Danny | All |
| README and repo tidy | Danny | All |

# 13. Demo script (4 minutes + 2 minutes for judges)

| Time | Speaker | On screen |
|---|---|---|
| 0:00 to 0:30 | Izzy | The problem: patients know they tripped. 29.5% of trials measure blinding; many fail above 90%. FDA July 2026 names it the central threat. |
| 0:30 to 1:30 | Danny | Pre-dosing: paste a draft protocol. Risk score, ranked fixes, FDA checklist, drafted blinding section. Approve a fix and watch it re-check. 260 applicants screened to a shortlist; flagged free-text disclosures shown with quotes. |
| 1:30 to 2:40 | Callum | Press play. Live blinding chart per site. Site B's raters drift; the agent explains and proposes remote raters; approve; drift stops. Show a debrief turned into structured events. |
| 2:40 to 3:30 | Izzy | Post-dosing: researcher asks a leading question; the bias gate rewrites or blocks it. Close-out FDA blinding report: reported vs expectancy-adjusted effect, with every decision traceable. |
| 3:30 to 4:00 | Danny | What it means: one standard, every trial comparable, a dataset that improves every new trial. Next step: pilot with one CRO. |
| 4:00 to 6:00 | Judges | They edit the protocol, play a patient, or submit a question. Everything is live. |

# 14. Glossary

| Term | Plain meaning |
|---|---|
| Blinding | Keeping secret who got the real drug |
| Functional unblinding | The drug's effects give the secret away |
| Expectancy | Feeling or scoring better because you believe you got the drug |
| Guess accuracy, u | Share of correct 'which group?' guesses. u = 2 x accuracy - 1, so 0 is a coin flip and 1 means everyone knew |
| Bang blinding index | Standard blinding score per arm, from -1 to 1, where 0 is ideal |
| Central blinded rater | The person scoring the patient is remote, never sees the session, and does not know the arm |
| Stratified randomisation | Randomising separately within groups (for example, experienced vs naive), so both arms get the same mix |
| MADRS | A clinician-scored depression scale, 0 to 60, higher is worse |
| Primary endpoint and lock | The one outcome the trial promised to test; locking it means no more changes before analysis |
| Hedges' g | Effect size in standard-deviation units, comparable across scales |
| Meta-regression | A line fitted across many trials, one dot per trial |
| CRO | Contract research organisation: companies that run trials for drug makers |
| Synthetic data | Simulated data, labelled as such, never presented as real-world performance |